

EXPERIMENT – 31

Implementation of CMAC Subkey Generation

CODE:

```
#include <stdio.h>
```

```
void leftShift(unsigned char in[], unsigned char out[], int n)
```

```
{
    int carry = 0;

    for (int i = n - 1; i >= 0; i--)
    {
        int newCarry = (in[i] & 0x80) ? 1 : 0;
        out[i] = (in[i] << 1) | carry;
        carry = newCarry;
    }
}
```

```
void printHex(unsigned char a[], int n)
```

```
{
    for (int i = 0; i < n; i++)
        printf("%02X ", a[i]);
    printf("\n");
}
```

```
int main()
```

```
{
    // Example value L = E(K, 0)
```

```
unsigned char L[8] = {0x80, 0x12, 0x34, 0x56,  
                      0x78, 0x9A, 0xBC, 0xDE};
```

```
unsigned char K1[8], K2[8];
```

```
// Constants for 64-bit block size
```

```
unsigned char Rb = 0x1B;
```

```
int msb = (L[0] & 0x80);
```

```
leftShift(L, K1, 8);
```

```
if (msb)
```

```
    K1[7] ^= Rb;
```

```
msb = (K1[0] & 0x80);
```

```
leftShift(K1, K2, 8);
```

```
if (msb)
```

```
    K2[7] ^= Rb;
```

```
printf("L = ");
```

```
printHex(L, 8);
```

```
printf("K1 = ");
```

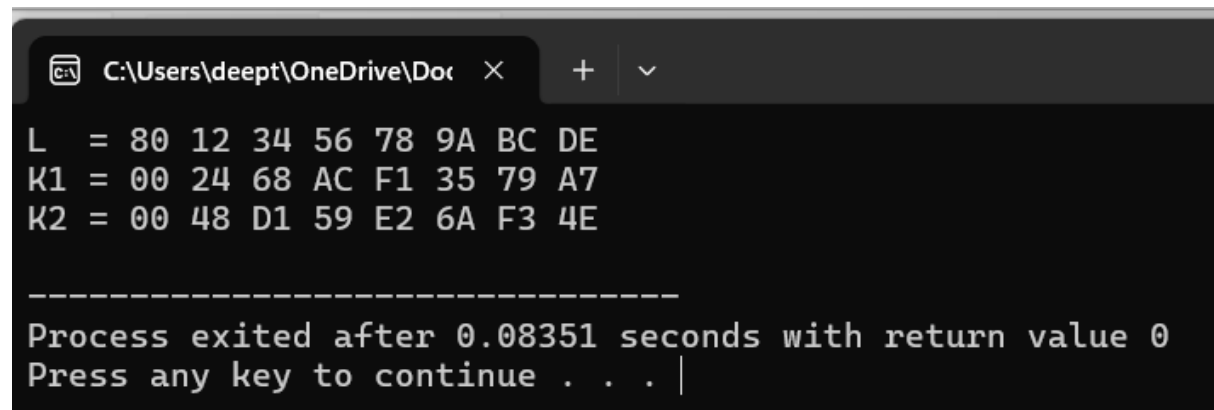
```
printHex(K1, 8);
```

```
printf("K2 = ");
```

```
    printHex(K2, 8);

    return 0;
}
```

OUTPUT:



```
C:\Users\deept\OneDrive\Doc >
L  = 80 12 34 56 78 9A BC DE
K1 = 00 24 68 AC F1 35 79 A7
K2 = 00 48 D1 59 E2 6A F3 4E

-----
Process exited after 0.08351 seconds with return value 0
Press any key to continue . . . |
```